

TITLE : SMART MEDICINE DISPENSER AND ALARMING KIT

ABSTRACT

The Smart Medication Alarming & Dispensing Kit presents an innovative approach to healthcare, particularly focusing on improving medication adherence for elderly users and patients with chronic illnesses. The device addresses environmental concerns by promoting sustainability in healthcare products. While traditional medication dispensers may lack eco-friendly components, our project emphasises the importance of using biodegradable and sustainable materials wherever possible. The design incorporates eco-friendly alternatives like 3D printed components made from biodegradable plastics, which align with the broader industry shift toward minimising environmental impact. By leveraging cutting-edge technology such as voice commands, IoT integration, and remote monitoring, this product balances functionality with environmental responsibility. Through this project, we aspire to create a sustainable, user-friendly, and effective medication management solution. Our emphasis on both technological advancement and ecological mindfulness signifies a commitment to fostering a harmonious relationship between healthcare innovations and environmental stewardship.

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THE FIELD OF INVENTION (PACKAGING INDUSTRY)

The invention lies in innovative advancements within the healthcare and assistive technology industries. Specifically, it addresses medication management and patient safety through automated dispensing, cloud connectivity, and user-friendly features. This invention enhances medication adherence for elderly users and individuals with chronic illnesses, aiming to improve patient outcomes and reduce manual errors.

BACKGROUND OF THE INVENTION

The Smart Medication Alarming & Dispensing Kit addresses the critical need for reliable medication management for elderly individuals and patients with chronic illnesses. Managing medication adherence is especially challenging for the ageing population, where missed or incorrect doses can lead to serious health issues. Traditional systems lack the necessary technology integration to support independent living effectively. This invention uses advanced automation, voice commands, and cloud connectivity to ensure precise and simplified medication dispensing. Servo motors and sensors enhance accuracy and minimise human error, while the user-friendly design is scalable and customizable for various medical needs. Real-time monitoring and IoT integration enable remote access for caregivers, ensuring a hands-free, hygienic, and efficient solution that bridges gaps in current medication management practices.

SUMMARY OF INVENTION

SmartMedDispenser: Smart medicine dispenser and alarming kit An advanced healthcare device developed to optimise medication adherence and safety, especially for the elderly and patients with chronic illnesses.

SPECIFICATION

- The Smart Medication Alarming & Dispensing Kit utilises a well-balanced combination of advanced hardware and software components to ensure effective and user-friendly medication management. Key materials and elements include a lightweight and modular 3D-printed frame, enhancing durability and ease of customization.
- Servo motors are integrated for precise control of compartment mechanisms, allowing for accurate dispensing of medications. This ensures the system operates smoothly and reliably, minimising the risk of dosage errors.
- The E3PS3 sensor is employed to detect medication presence and confirm accurate dispensing, preventing over- or under-dosing. This sensor plays a critical role in maintaining safe and correct medication administration.
- The central processing unit is an Arduino microcontroller, which manages inputs from the sensors and operates the servo motor efficiently. This robust setup ensures the seamless functioning of the entire system.
- Software features include cloud connectivity for real-time monitoring, enabling caregivers to receive alerts and updates about missed doses of medication adherence. The integration of a voice command system and a user-friendly interface makes the dispenser accessible to elderly users.
- The combination of automation, cloud-based monitoring, and modular design makes the device environmentally friendly and sustainable, as it reduces waste associated with missed doses and enables easy upgrades or repairs.

DESCRIPTION:

The Smart Medicine Dispenser and Alarming Kit project integrates a bioplastic with a pH indicator, which is crafted from natural materials like cornstarch and banana starch, with glycerin and glycerol as plasticizers, and gelatin as a solidifier. The addition of red cabbage extracts introduces a unique feature: the bioplastic changes colour based on its surrounding pH levels. Under acidic conditions, it displays vibrant red and pink hues, while in neutral environments, it turns purple or violet. In basic conditions, the colour shifts to blue and green. This bioplastic is not only biodegradable and eco-friendly but also highly versatile, making it ideal for educational purposes, environmental monitoring, and applications requiring pH sensitivity. It represents a major leap forward in the development of sustainable materials for use in the plastic and packaging industries.



Figure 1



Figure 2

WE CLAIM

1. Claim: The Smart Medicine Dispenser with real-time alerts and medication reminders ensures timely medication adherence, providing an intuitive, user-friendly experience for elderly users and patients with limited technical knowledge.
2. Claim: The Smart Medicine Dispenser is designed for convenience and safety, allowing users to open and close the dispenser through a mobile app, with an audible beep to alert users when it is time to take their medication.
3. Claim: The Smart Medicine Dispenser integrates cloud-based monitoring, ensuring that caregivers and family members are notified instantly through real-time alerts if a user misses a dose or requires assistance.
4. Claim: The Smart Medicine Dispenser offers a significant advancement in medication management, reducing the risk of missed doses and improving medication adherence in patients, particularly those with chronic conditions or complex medication regimens.
5. Claim: The Smart Medicine Dispenser's eco-friendly design incorporates a 3D-printed frame, offering an environmentally conscious alternative to traditional, non-sustainable materials.

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